

PIPESTONE, MN, USA

WMO: 726566

Lat: 43.983N

Lon: 96.300W

Elev: 1736

StdP: 13.80

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04965

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-8.0	-18.0	2.2	-10.2	-14.6	2.7	-7.1	29.1	16.3	26.5	17.9	7.7	300	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	89.8	73.8	86.3	72.6	83.6	70.9	77.7	84.7	75.9	83.0	74.0	80.6	12.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	141.7	82.2	73.2	131.5	80.4	72.0	126.4	79.3	42.7	85.1	40.6	83.2	38.8	80.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	23.6	20.5	-19.5	94.1	6.3	3.1	-24.1	96.3	-27.7	98.1	-31.2	99.9	-35.8	102.1		
(5)			-20.3	80.6	6.3	2.8	-24.8	82.6	-28.5	84.3	-32.0	85.9	-36.5	87.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	15.7	19.9	31.7	45.5	57.4	68.0	72.4	69.3	62.2	48.6	33.5	20.6		
	DBStd	22.25	12.97	12.95	12.82	10.01	8.84	6.73	5.67	5.46	8.21	9.83	11.40	12.12		
	HDD50	4280	1063	842	577	197	32	0	0	0	7	145	504	913		
	HDD65	7737	1528	1262	1032	587	267	44	9	22	149	514	945	1378		
	CDD50	2653	0	0	11	63	261	541	693	600	373	102	9	0		
	CDD65	635	0	0	0	3	31	135	238	157	64	7	0	0		
	CDH74	5457	0	0	2	50	349	1208	2104	1132	528	83	1	0		
	CDH80	1569	0	0	0	11	100	371	674	271	123	19	0	0		
(14) Wind	WSAvg	9.2	9.8	10.2	10.3	10.8	10.5	8.8	7.2	6.7	8.2	9.3	9.7	9.6		
(15) Precipitation	PrecAvg	25.7	0.6	0.6	1.5	2.6	3.3	3.9	3.2	3.0	3.1	2.0	1.3	0.6		
	PrecMax	38.6	1.7	2.5	4.1	7.6	8.6	9.8	7.8	8.3	10.2	6.3	4.4	2.0		
	PrecMin	9.1	0.0	0.0	0.1	0.4	0.2	1.3	1.0	0.6	0.4	0.0	0.0	0.0		
	PrecStd	5.7	0.5	0.5	1.0	1.7	2.1	2.2	1.9	1.7	2.0	1.6	1.1	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	55.1	70.2	81.1	89.8	91.3	92.7	90.3	88.2	82.2	68.4	51.5		
		MCWB	38.4	45.2	54.8	58.7	65.2	72.8	76.8	76.6	69.8	62.4	53.6	45.0		
	2%	DB	38.5	46.2	62.0	72.7	82.0	88.1	90.0	85.9	83.5	74.7	61.2	44.5		
		MCWB	35.0	40.5	52.2	55.0	62.9	70.9	76.1	75.6	70.2	59.6	50.2	39.1		
	5%	DB	36.0	40.8	54.9	66.4	77.9	84.0	87.4	82.3	80.8	69.8	54.6	38.6		
		MCWB	33.9	37.3	47.9	52.0	61.5	69.1	75.1	72.3	67.2	57.7	46.6	34.8		
	10%	DB	33.7	36.7	48.5	62.8	72.8	81.2	83.7	80.9	76.5	63.9	48.4	36.1		
		MCWB	32.0	34.0	43.7	50.3	59.0	67.5	72.7	70.8	65.2	54.7	42.6	33.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	48.7	58.1	61.7	70.8	77.6	81.4	80.5	74.9	67.9	56.8	45.5		
		MCDB	44.8	54.3	64.2	74.5	81.7	86.8	87.6	86.4	82.6	76.1	64.2	48.0		
	2%	WB	35.9	41.2	53.0	57.8	66.8	75.1	78.6	77.0	72.4	63.3	51.5	38.8		
		MCDB	38.2	45.6	59.3	67.9	77.0	82.9	85.2	83.2	79.8	68.4	59.0	43.0		
	5%	WB	33.7	37.3	48.8	54.7	64.1	72.6	76.8	75.0	70.0	59.4	47.1	35.7		
		MCDB	35.5	40.2	55.8	63.5	72.5	79.9	84.0	81.0	77.0	66.9	53.8	38.2		
	10%	WB	31.6	34.3	43.7	52.1	61.4	70.0	74.8	72.7	67.6	55.6	43.2	33.4		
		MCDB	33.4	36.7	49.1	60.5	70.3	78.4	81.9	78.2	74.5	63.0	48.7	35.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	17.0	16.6	19.8	19.8	18.9	19.2	18.5	20.8	19.7	17.5	15.1		
		MCDBR	18.9	21.2	25.6	29.2	26.5	22.7	21.0	20.4	24.1	27.0	25.5	19.6		
		MCWBR	15.4	15.9	16.6	15.3	13.0	10.8	10.9	11.9	12.2	14.7	15.9	15.2		
	5% WB	MCDBR	17.2	19.6	24.5	24.2	22.0	19.4	18.9	18.5	20.0	22.4	23.9	18.2		
		MCWBR	14.5	15.2	16.6	15.0	12.5	11.6	11.3	12.1	12.3	13.9	15.8	14.8		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.278	0.310	0.363	0.380	0.391	0.399	0.402	0.356	0.319	0.285	0.267		
	taud		2.449	2.417	2.424	2.304	2.333	2.359	2.328	2.292	2.401	2.487	2.502	2.445		
	Ebn at Noon		281	293	296	285	283	279	275	270	275	274	268	266		
	Edh at Noon		23	28	32	39	39	38	39	39	32	26	21	21		
(44) All-Sky Solar Radiation	RadAvg		596	931	1214	1525	1761	1966	2105	1738	1376	922	626	464		
	RadStd		59	87	109	144	130	131	77	137	102	113	48	41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page